

3. (a) The list below shows part of the reactivity series.

sodium
aluminium
(carbon)
tin
copper
silver

- (i) Tin is extracted from its ore by heating with carbon. Aluminium is extracted from its ore using a different method. Give the name of the method used to extract aluminium. [1]

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- (ii) The equation shows the extraction of tin from tin oxide using carbon.



Tick (✓) the box next to the correct statement. [1]

Carbon is reduced

☐

Tin is oxidised

☐

Tin oxide is reduced

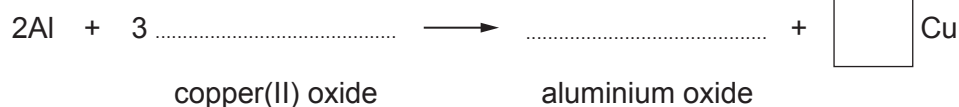
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Carbon dioxide is oxidised

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- (iii) When aluminium and copper(II) oxide are heated together, aluminium oxide and copper are formed.

Complete and balance the equation for this reaction. [3]



- (b) A teacher wanted to find out the position of four metals **A**, **B**, **C** and **D** in the reactivity series.

She heated each metal in turn with oxides of the other three. The results were as follows.

A reduced the oxide of **C**

B reduced the oxide of **A**

B reduced the oxide of **C**

D reduced the oxide of **B**

Place the metals in order of reactivity.

[2]

Most reactive

.....

.....

Least reactive

